Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.23	20.25	
worker nodes in service, mean	6.00	5.05	-15.8%
worker nodes in service, min	6.00	5.00	-16.7%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	1.70	-15.7%
power (W), mean	629	566	-10.0%
power (W), peak	654	622	-4.8%
energy (Wh)	212	191	-10.0%
energy, parked workers still on at idle power (Wh)	212	215	+1.3%
CPU used (cores), mean	0.906	1.09	+20.4%
CPU allocatable (cores), mean	24.00	20.21	-15.8%
utilisation (used / allocatable)	0.038	0.054	+42.9%
energy per core-hour (Wh)	694	519	-25.3%
node-hours per core-hour	6.62	4.63	-30.0%
pending pods, pod-minutes	0.000	0.250	new
pending pods, peak	0.000	1.00	new
HPA replicas, mean	8.92	9.22	+3.3%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	1.33	0.783	-41.2%
requests timed	1,040	1,115	+7.2%
time over the response line (% of samples)	0.769	0.000	-100.0%
response time (ms), mean	155	76.87	-50.4%
response time (ms), median	128	69.50	-45.7%
response time (ms), 95th percentile	327	111	-66.0%
response time (ms), 99th percentile	475	139	-70.7%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.2350	5.3372	-26.2%	[-3.6475, -0.1446]	GAIN
kwh_per_core_hour	0.7555	0.5915	-21.7%	[-0.3345, +0.0106]	EVEN, no significant difference
utilisation	0.0373	0.0537	+44.2%	[+0.0031, +0.0310]	GAIN
pending_pod_minutes_per_hour	0.0000	0.7680	n/a	[+0.0000, +2.3040]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	3.8157	2.3164	-39.3%	[-6.1013, +2.3535]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.358 0.392 0.416 0.437 0.457 0.473 0.488 0.501 0.513 0.523 0.532 0.536 0.54 0.544 0.549 0.553 0.556 0.559 0.56 0.56
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	617	617	4.25	5.00	0.000	0.000	0.579	0.572

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	5.00	620	545	5.75	6.00	0.000	0.000	0.630	0.638
2	6.00	5.00	618	547	6.00	7.00	0.000	0.000	0.610	0.692
3	6.00	5.00	624	551	6.25	7.00	0.000	0.250	0.775	0.792
4	6.00	5.00	634	569	9.00	9.75	0.000	0.000	1.04	1.27
5	6.00	5.00	632	565	9.00	10.00	0.000	0.000	0.964	1.15
6	6.00	5.00	636	570	9.00	10.00	0.000	0.000	1.05	1.29
7	6.00	5.00	647	592	9.75	10.00	0.000	0.000	1.39	1.85
8	6.00	5.00	645	590	10.00	10.00	0.000	0.000	1.31	1.85
9	6.00	5.00	645	588	10.00	10.00	0.000	0.000	1.33	1.79
10	6.00	5.00	651	576	10.00	10.00	0.000	0.000	1.52	1.46
11	6.00	5.00	618	547	10.00	10.00	0.000	0.000	0.617	0.670
12	6.00	5.00	619	546	10.00	10.00	0.000	0.000	0.625	0.648
13	6.00	5.00	626	552	10.00	10.00	0.000	0.000	0.808	0.830
14	6.00	5.00	633	570	10.00	10.00	0.000	0.000	1.02	1.30
15	6.00	5.00	638	572	10.00	10.00	0.000	0.000	1.10	1.33
16	6.00	5.00	632	571	10.00	10.00	0.000	0.000	0.973	1.32
17	6.00	5.00	626	556	10.00	10.00	0.000	0.000	0.811	0.913
18	6.00	5.00	620	547	10.00	10.00	0.000	0.000	0.634	0.686
19	6.00	5.00	616	546	10.00	10.00	0.000	0.000	0.551	0.651
20		5.00		549		10.00		0.000		0.722

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.